House Prices Prediction With Advanced Regression Techniques : The Comparison Of Random Forest ,XgBoost And PCA

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1 Introduction

Housing prices are one of the most important indicators of economic stability and social well-being. The housing market is influenced by many factors, including location, property size, number of rooms, construction quality, and macroeconomic conditions. In recent years, the availability of large-scale real estate datasets and advancements in machine learning techniques have provided more accurate and data-driven approaches to housing price prediction. As a result, predictive modeling has become a crucial tool for understanding housing market dynamics and supporting decision-making processes.

Problem Definition:

Accurately predicting housing prices remains a challenging task due to the complex and nonlinear relationships between property attributes and market value. Traditional valuation methods often rely on expert opinion and limited comparisons, leading to subjective and inconsistent results. This creates uncertainty for buyers, sellers, and investors who need reliable price predictions in a rapidly changing market environment.

Purpose and Scope of the Study:

The aim of this study is to develop a predictive model for housing prices using supervised machine learning techniques. The study focuses on the analysis of a structured housing dataset containing various property attributes and their corresponding sales prices. Within this research, data preprocessing, feature selection, and model training processes are applied to evaluate the effectiveness of different regression-based approaches in predicting housing prices.

Scientific or Practical Significance:

From a practical standpoint, accurate home price prediction models can help real estate professionals, investors, and potential buyers make informed decisions. Scientifically, this study contributes to understanding how machine learning algorithms capture

relationships between housing properties and prices, as well as highlighting the importance of data quality and feature engineering in predictive modeling. The findings can serve as a reference for future work in the fields of real estate analytics and applied machine learning.

2 Aim Of The Project

The main aim of this project is to predict house prices by applying supervised machine learning regression techniques on a structured housing dataset. The project specifically aims to analyze the relationship between property features and sale prices, and to evaluate how accurately different regression models can predict house prices.

The objectives of Project are below:

- To analyze the data and clean the missing parts of data for modeling
- To apply 2 regression models at least for house price's prediction
- To evaluate and compare the performance of these models by checking RMSE and R2 score
- To decide which model makes more acccurate prediction based on the obtained results

By achieving these objectives, the project aims to demonstrate the effectiveness of machine learning methods in solving real-world regression problems related to the housing market.

3. Literature Review

3.1 Summary of Previous Studies

Housing price prediction has been extensively studied due to its economic significance for homeowners, investors, and policy-makers. While early research relied heavily on traditional statistical methods such as Hedonic Pricing Models [1], recent

advancements have shifted towards Machine Learning (ML) techniques to capture complex, non-linear relationships in housing data [2].

The literature demonstrates a clear evolution from simple linear regression models to sophisticated ensemble methods. Studies consistently show that tree-based ensemble algorithms, particularly Random Forest and Gradient Boosting, outperform traditional econometric approaches when applied to high-dimensional datasets [3]. However, a recurring challenge cited in the literature is the trade-off between model complexity and interpretability, often leading to overfitting in small-to-medium-sized datasets [4].

3.2 Datasets Used in Research

Previous studies have relied on various datasets, ranging from the classic Boston Housing dataset to more complex modern datasets.

- **The Ames Housing Dataset**: Compiled by Dean De Cock [5], this dataset (which is used in this study) is considered a gold standard in recent academic research. With approximately 1460 samples and 79 explanatory variables, it offers a richer feature set than older datasets, allowing for more in-depth feature engineering and outlier analysis.
- **Regional Datasets:** Other studies utilize proprietary data from real estate platforms (e.g., Zillow), which offer vast quantities of data but often lack detailed property attributes like "Garage Quality" or "Basement Finish," which are critical for precise valuation.

3.3 Methodological Approaches

A wide range of algorithms has been applied to this regression problem:

- **Linear & Regularized Models:** Ridge and Lasso regression are frequently used as baselines. While they effectively handle multicollinearity through regularization, they often fail to capture non-linear patterns inherent in real estate data [6].

- **Random Forest:** Proposed by Breiman [7], this bagging technique is widely praised in the literature for its robustness against overfitting and its ability to handle mixed data types without extensive scaling.
- **XGBoost (Extreme Gradient Boosting**): Chen and Guestrin [8] introduced XGBoost, which has become the state-of-the-art method in housing prediction competitions. Literature highlights its superior performance due to its regularization parameters ($L1/L2$), which explicitly control model complexity—a feature central to our study's methodology.
- **Dimensionality Reduction (PCA):** Some studies apply Principal Component Analysis (PCA) to reduce feature space. However, results are mixed; while it speeds up training, several researchers note that in housing datasets, removing features often leads to information loss and reduced accuracy [9].

3.4 Strengths and Limitations of Existing Literature

The greatest strength of current research is the demonstration of ensemble methods' superiority. However, several limitations remain:

- **Overfitting:** Many studies achieve high accuracy on training data but fail to generalize well due to a lack of rigorous cross-validation or regularization.
- **Data Quality Neglect:** A significant portion of research focuses solely on algorithm selection, neglecting the impact of advanced data cleaning (e.g., imputation strategies) on model performance.

3.5 Contribution of This Study

This project contributes to the literature by conducting a systematic comparison of Random Forest and Regularized XGBoost on the Ames dataset. Unlike studies that focus only on accuracy, this research specifically addresses the "Overfitting" problem by implementing a tuning strategy that reduced the generalization gap. Furthermore, we

experimentally demonstrate the inefficiency of PCA for this specific domain, reinforcing the finding that high-dimensional feature retention is crucial for accurate real estate valuation.

4 Materials

4.1 Dataset Description

The dataset used in this study is a structured housing dataset designed to predict housing sales prices[5]. Because it contains detailed information on the physical, locational, and qualitative characteristics of homes (figure 1.1) ,it is suitable for supervised machine learning regression tasks..

Total Sample Size:

The dataset consists of 1,460 samples, each representing a single residential property. Each sample includes multiple descriptive features along with the target variable representing the final sale price of the home

--- First 5 Row ---

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	...	PoolArea	PoolQC	Fence
0	1	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN
1	2	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	...	0	NaN	NaN
2	3	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN
3	4	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN
4	5	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	...	0	NaN	NaN

5 rows × 81 columns

. **Figure 1.1 df.head()**

Dataset Source:

The dataset is publicly available and was obtained from Kaggle, originally compiled from real estate records of residential homes in Ames, Iowa, USA. It is widely used in academic studies and machine learning competitions due to its rich feature set and real-world relevance.

Data Type:

The dataset is tabular in nature and includes both:

- **Numerical features** (e.g., lot area, living area, year built, number of rooms)
- **Categorical features** (e.g., neighbourhood, zoning classification, building type, exterior quality)

The target variable, SalePrice, is a continuous numerical value, making the problem a regression task.

Class Distribution:

Since the task is formulated as a regression problem, the dataset does not contain discrete class labels. Instead, the target variable (SalePrice) is continuously distributed across a wide price range. Some studies arbitrarily convert prices into categorical ranges (e.g., low, medium, high) for classification purposes; however, in this study, the original continuous price values have been preserved to maintain estimation accuracy.

Annotation Format:

No manual annotation process is involved in this dataset. All feature values and target labels are derived directly from structured real estate records. Therefore, no annotation format is applicable, as the dataset does not include image data, segmentation masks, or expert-labeled classes.

4.2 Pre-Processing Steps

Figure 1.2 Missing values heatmap(before cleaning)

There is a heatmap of our dataset above which shows us the missing values (figure 1.2). To prepare this uncleaned data for machine learning models, the following steps were implemented:

Noise Reduction(Feature Selection):

Columns with too many missing data (over 90%), such as PoolQC, Alley, and Fence, were considered "noise" and removed from the dataset to prevent model performance. The Id column was also removed from the dataset as it did not provide a predictive value an it increased the model's error rate.

Advanced Imputation(Handling Missing Data):

- **Regression Imputation:** For the LotFrontage column, simple mean replacement was avoided to preserve variance. Instead, a Linear Regression model was trained using 1stFlrSF (First Floor Square Footage) to predict and fill the missing frontage values

accurately.(It was chosen 1stFlrSF as it has the highest correlation with LotFrontage)(figure 1.3).

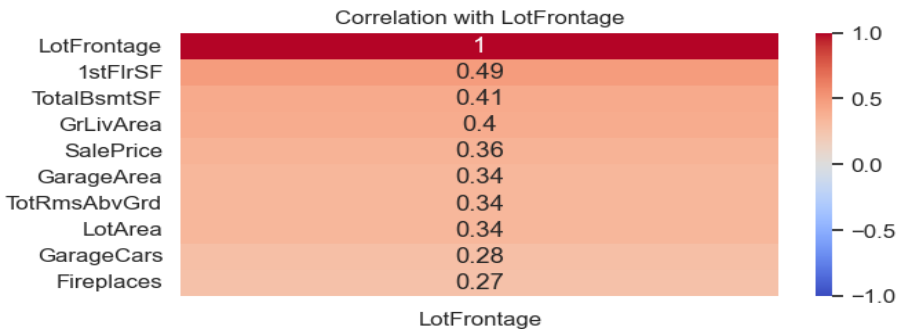

Figure 1.3 Correlation Matrix of LotFrontage

- **Logic-Based Imputation:** For features like Garage and Basement, missing values were interpreted as the absence of that feature. Therefore, categorical missing values were filled with "None" and numerical ones with "0". We also fill GarageYearBlt with the year of when the house was build . Otherwise our model could consider the GarageYearBuilt 0 and would be confused(figure 1.4).

```
for col in ['GarageType', 'GarageFinish', 'GarageQual', 'GarageCond']:
    df[col] = df[col].fillna('None')

df['GarageYrBlt'] = df['GarageYrBlt'].fillna(df['YearBuilt'])

for col in ['BsmtQual', 'BsmtCond', 'BsmtExposure', 'BsmtFinType1', 'BsmtFinType2']:
    df[col] = df[col].fillna('None')

df = df.dropna(subset=['Electrical'])
df = df.dropna()
```

Figure 1.4

Encoding:

- **Label Encoding:** It is useful for ordinal data where order matters . Our model can't understand strings. So , we convert them into numbers by considering their importance.(ExterQual)
- **One-Hot Encoding:** Applied to nominal data (e.g., Neighborhood) it is used to convert categories into binary vectors . When there is no concern about importance ,

we number those features as 0 and 1 where “0” means it does not have and “1” means it has that features.

You can see the implementation of encoding in my code below(figure 1.5).

```
ordinal_cols = [
    'ExterQual','ExterCond','BsmtQual','BsmtCond','KitchenQual','HeatingQC',
    'Functional','GarageFinish','BsmtExposure','BsmtFinType1','BsmtFinType2',
    'GarageQual','GarageCond'
]
obj_cols = df.select_dtypes(include=['object']).columns
for col in ordinal_cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col].astype(str))
nominal_cols = [c for c in obj_cols if c not in ordinal_cols]
df = pd.get_dummies(df, columns=nominal_cols, drop_first=True)
```

. **Figure 1.5 Implementation of Encoding part**

Normalization:

Because Our data such as lotArea and YearBlt has a huge difference , Our model may think that the feature which has the higher number has higher importance. To prevent this, we use scaler and make average 0 and standard deviation 1.

5 Methods

5.1 Data Splitting

Train/Validation/Test split

Our dataset is divided into two parts as %80 of train and %20 of test.

- **Train:** Used to teach and train my model.
- **Test:** Used to test the final performance of our model. It is not included in the train part to avoid overfitting.

It hasn’t been used validation part as We do not have too many sample in our dataset.

K-Fold cross-validation

5-fold cross-validation used here as we do not have too many data. With the benefit of it , we have seen how our model runs with the bigger dataset and detect if there is a overfitting .

5.2 Machine Learning

In our project, a few diffirent model applied and compared.

Applied Models:

Used models in our project are below:

- **Random Forest Regressor:**

  We selected this model as we have a lot of features in our dataset. Random Forest take independently every features as decision trees and calculate their averages. By doing this , we get more accurate results from the high-dimensional datasets.

- **XGBoost Regressor:**

  We selected this model due to it is good performance on complex but structured datasets.

  Additionaly , XGBoost works more sensitive and has more accuracy as it is a boosting model and learns from the previous branch.

Note: We reduced the depth of these models and used regulization for XGBoost to prevent overfitting. Our model learns from the noise and elaborated features, when we go to deep very detailed. To prevent the model from memorizing noise and overfitting, we do these things

- **Hybrid Model(PCA + XGBoost):** It was used to test if reducing dimensionality would increase the accuracy and training speed. But We have seen that it had a significant decrease in performance . From these knowledge, we can infer that almost every feature we have has an efficient contribution to our result

Hyperparameters:

Used to maximize performance and prevent overfitting.

```
rf_model = RandomForestRegressor(
    n_estimators=300,
    max_depth=20,
    min_samples_split=5,
    min_samples_leaf=2,
    max_features=None,
    random_state=42
)
```

```
xgb_model = XGBRegressor(
    n_estimators=2000,
    learning_rate=0.01,
    max_depth=4,
    subsample=0.7,
    colsample_bytree=0.7,
    reg_alpha=10,
    reg_lambda=10,
    gamma=0.5,
    random_state=42,
    n_jobs=-1
```

Training Details:

The training process was conducted on 80% of the dataset.

- **Strategy:**

  The gap between Train RMSE and Test RMSE was monitored to mitigate overfitting.

- **Validation: 5-Fold Cross-Validation**:

  It was applied during training to ensure model stability.

- **Environment:**

  Training was executed on a standard CPU using all available cores (n_jobs=-1).

Libraries Used:

The project was developed using Python. The used libraries are below;

- **Data Manipulation:** Pandas
- **Visualization:** Matplotlib, Seaborn
- **Machine Learning:** Scikit-Learn, XGBoost

The model performance was evaluated using RMSE (Root Mean Squared Error) to measure the average error magnitude and R^2 Score to explain the variance. We also looked at the Error rate(RMSE/mean)(table 1.1).

	Random Forest	XGBoost	RandomF + K-Fold	XGBoost + K-Fold	PCA + XGBoost
RMSE	21.303$	19.357$	31.823$	27.939$	26.417$
R^2	%91.87	%93.28	%83.97	%87.71	%87.49
Error Rate	%11.79	%10.72	%17.62	%15.47	%14.63

Table 1.1 Model Performance Summary

Table 1 summarizes the evaluation metrics for our models. As clearly seen, the XGBoost model is the most efficient approach, ac

It is important to note that the K-Fold Cross-Validation RMSE scores are higher than the single Test RMSE scores. This is not a negative result; on the contrary, it is expected. The K-Fold RMSE is higher because it evaluates the model on multiple, challenging subsets of the data to measure its true stability. This confirms that our model is robust and not just

successful on a single "lucky" data split.

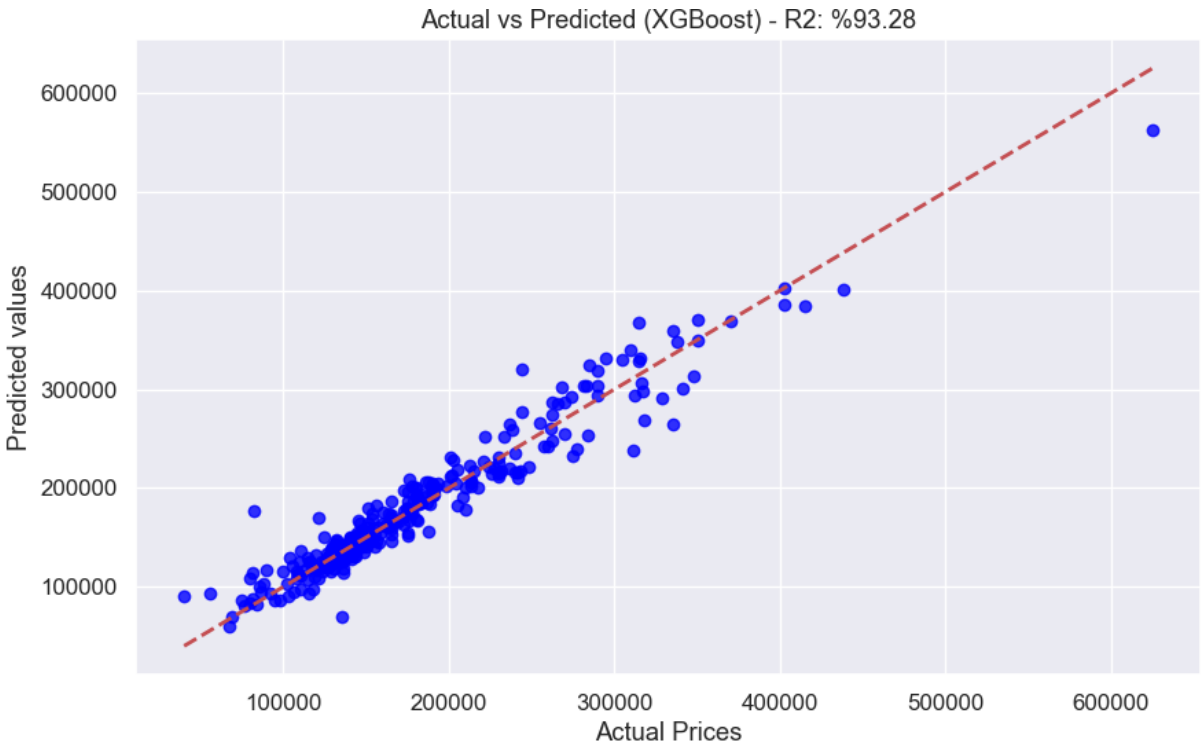

Figure 1.6 XGBoost

The scatter plot demonstrates a strong linear relationship between the predicted and actual values. The data points cluster tightly around the red reference line (ideal fit)(figure1.6)

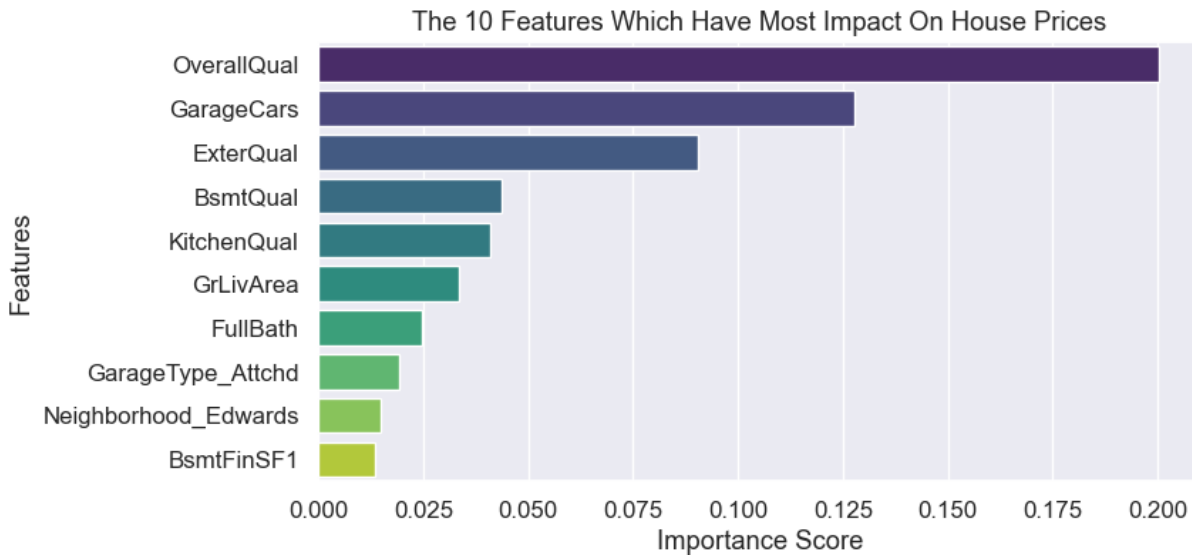

Figure 1.7 Feature Importance Table

Top 10 Features Influencing House Prices(figure 1.7). Interpretation: According to the XGBoost model, the most critical factor determining a house's price is 'OverallQual' (Overall Material and Finish Quality), followed by 'GrLivArea' (Living Area). This aligns with real-world expectations that quality and size are the primary drivers of value.

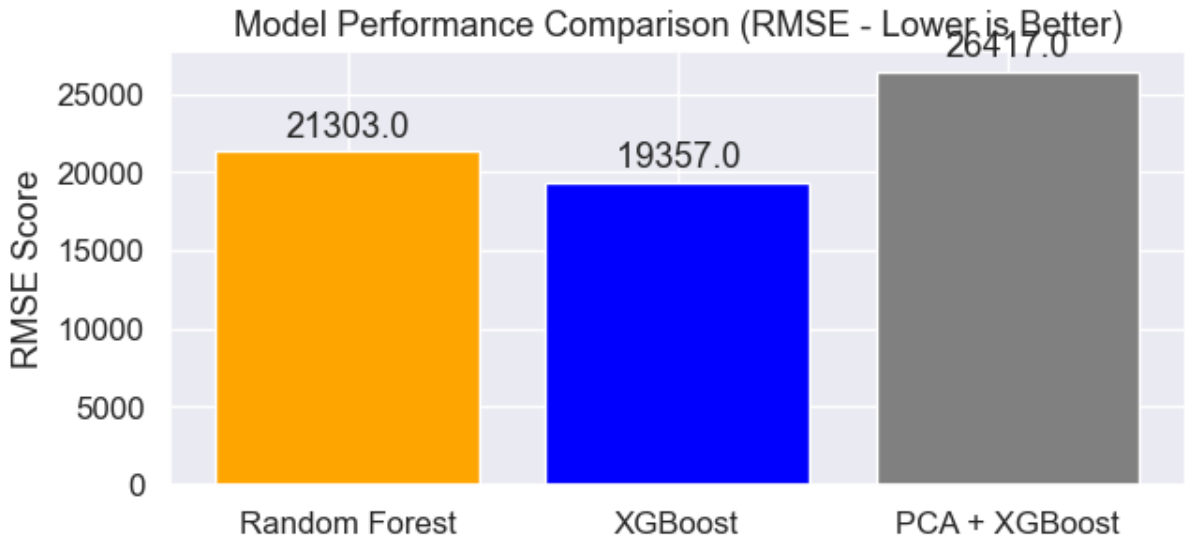

Figure 1.8 The comparison of the models

The comparison table of the models above shows us clearly why we have chosen XGBoost(figure 1.8).

7 Discussion

7.1 Interpretation of Model Performances:

The results show that the XGBoost model performed the best among all methods, achieving the lowest RMSE (19,357) and the highest R^2value (93.28%).

- The Random Forest model produced reliable results but had a higher error, approximately 2,000 units worse than XGBoost.
- In contrast, applying PCA led to a clear drop in performance, with the R^2value decreasing to around 79%. This indicates that most features in the dataset are informative and that reducing dimensionality removes important information needed for accurate price prediction.

7.2 Why XGBoost Performed Better?

XGBoost performed better than the other models because it improves predictions step by step by correcting previous errors. At first, the model suffered from overfitting, as the training error was much lower than the test error. This issue was reduced by applying regularization (L1 and L2) and limiting the tree depth. As a result, the model learned more general patterns instead of memorizing the training data and achieved better performance on unseen data.

7.3 Limitations of the Study

- **Dataset Size:**

The model was trained on a relatively small dataset (1460 entries). A larger dataset could further improve stability.

- **Outliers:**

Extreme outliers in house prices (e.g., very large luxury houses sold cheaply) negatively impacted the RMSE, as regression models are sensitive to outliers.

7.4 Compare & Insights For Future Work

The achieved RMSE value of 19,357 is competitive when compared to commonly reported results on the Ames Housing dataset, which typically range between 15,000 and 25,000 for advanced non-ensemble models. This indicates that the proposed approach performs well within established benchmarks.

For future work, model performance could be further improved by applying stacking or ensemble techniques, such as combining XGBoost with Lasso Regression. In addition, deep learning approaches, including neural networks, may be explored if a larger dataset becomes available.

8 Conclusion

In this project, we developed a reliable machine learning pipeline to predict house prices using the Ames Housing dataset. The study focused not only on model selection but also on careful data preprocessing, including regression-based imputation for missing values and effective feature engineering.

Through a comparison of Random Forest, XGBoost, and PCA-based approaches, several important findings were obtained. The XGBoost model achieved the best performance, with an RMSE of 19,357 and an R^2score of 93.28%. This shows that the model can explain most of the variation in house prices with high accuracy.

Another key outcome of this work was the successful reduction of overfitting. By applying L1 and L2 regularization and limiting tree depth, the model was able to generalize well to unseen data, as confirmed by cross-validation results.

The results also showed that dimensionality reduction using PCA reduced performance. This indicates that most features in the dataset contain valuable information and should not be removed. Feature importance analysis further revealed that overall house quality and living area are the most influential factors in determining house prices.

In conclusion, this study presents an accurate and practical house price prediction model. The results highlight the importance of proper data preprocessing, thoughtful feature selection, and careful model tuning for achieving strong predictive performance in real-world applications.

9 References

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